

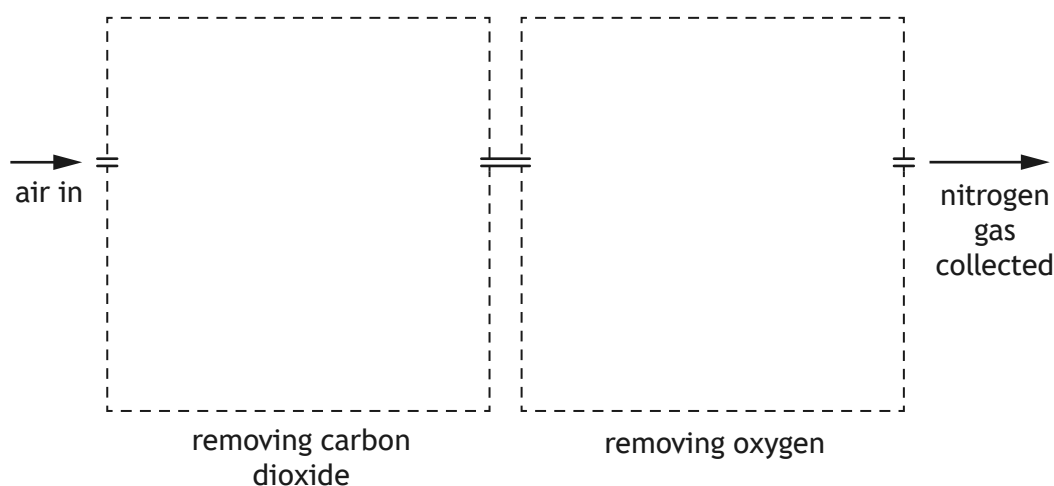
4. Nitrogen is a stable element and only reacts with a few other elements.

- (a) Explain, using bond enthalpy values, why the element nitrogen is so unreactive.

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- (b) A laboratory preparation of nitrogen gas involves separating nitrogen from all of the other gases in the air and then collecting it.

Air is bubbled through potassium hydroxide solution to remove carbon dioxide. The remaining gases are passed over heated copper to remove oxygen before the remaining nitrogen is collected.



Complete a labelled diagram for the preparation of a sample of nitrogen by drawing suitable apparatus for carrying out this preparation.

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(An additional diagram, if required, can be found on *page 38*.)

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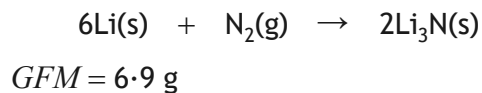
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4. (continued)

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- (c) Nitrogen can react with lithium at room temperature to form the compound lithium nitride, Li_3N .

- (i) A scientist prepared a sample of lithium nitride by reacting 0.9 litres of nitrogen gas with 0.5 g of lithium.



Determine, by calculation, which of the reactants was in excess.

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Take the volume of 1 mole of nitrogen gas to be 24 litres.

(Clearly show your working for the calculation.)

- (ii) Lithium nitride reduces copper(I) ions to copper atoms.

Write the ion-electron equation to show the reduction of copper(I) ions.

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- (iii) Lithium nitride is ionic.

State the term used to describe the structure of solid ionic compounds like lithium nitride.

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4. (continued)

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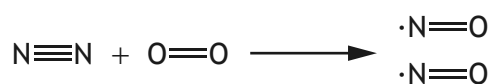
- (d) During thunderstorms, nitrogen can react with oxygen to form different compounds.

Nitrogen and oxygen can react to form the free radical molecule nitrogen monoxide, $\cdot\text{N}=\text{O}$.

- (i) State what is meant by the term free radical.

1

- (ii) The equation for this reaction is



The enthalpy change, ΔH , for this reaction is $+91 \text{ kJ mol}^{-1}$.

Use this data and the bond enthalpy values shown in the data booklet to calculate the bond enthalpy, in kJ mol^{-1} , of the nitrogen to oxygen double bond in nitrogen monoxide.

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- (iii) Nitrogen monoxide free radicals can react with hydroxyl free radicals to form a molecule of nitrous acid, HNO_2 .

- (A) Name the type of reaction that occurs when two free radicals join together.

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- (B) Draw a possible structure for the HNO_2 molecule.

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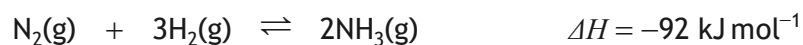


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4. (continued)

- (e) Nitrogen can also react with hydrogen in the Haber process to form ammonia.

- (i) The equation to produce ammonia is shown.



Explain fully the effect of decreasing the reaction temperature on the yield of ammonia.

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- (ii) Ammonia can be reacted to produce the explosive nitroglycerin. A small shock or physical bump can make the nitroglycerin explode. The reaction that takes place is shown.



- (A) Balance this equation.

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- (B) Suggest why a small shock or bump can cause nitroglycerin to react.

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